

The image shows a Wireshark packet capture analysis of an ARP request. The packet list on the left shows a packet from 192.168.1.10 to 192.168.1.1. The packet details pane on the right shows the Ethernet II header with source MAC 08:00:27:00:00:00 and destination MAC 08:00:27:00:00:00. The ARP section shows the request for the IP address 192.168.1.1.

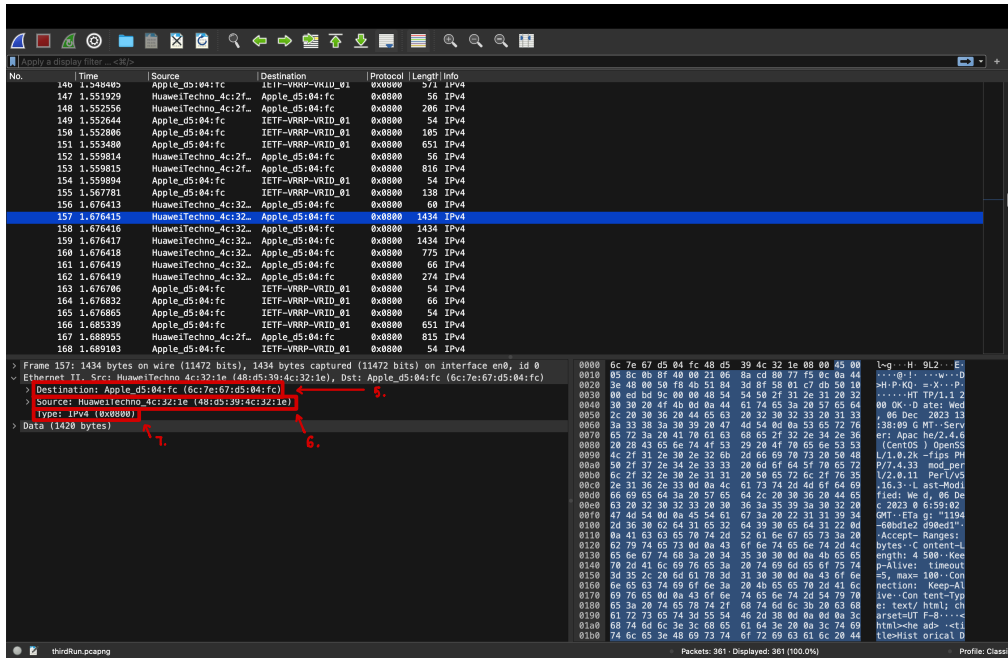
Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
112	1.351967	HuaweiTechno_4c:32:1e	Apple_d5:84:fc	0x8000	814	IPv4
113	1.352192	Apple_d5:84:fc	ITF7-VRRP-VRID_01	0x8000	54	IPv4
114	1.359949	Apple_d5:84:fc	ITF7-VRRP-VRID_01	0x8000	138	IPv4
115	1.362574	HuaweiTechno_4c:32:1e	Apple_d5:84:fc	0x8000	66	IPv4
116	1.376070	HuaweiTechno_4c:32:1e	Apple_d5:84:fc	0x8000	66	IPv4
117	1.376711	HuaweiTechno_4c:2f:1e	Apple_d5:84:fc	0x8000	66	IPv4
118	1.378939	Apple_d5:84:fc	ITF7-VRRP-VRID_01	0x8000	54	IPv4
119	1.371843	Apple_d5:84:fc	ITF7-VRRP-VRID_01	0x8000	54	IPv4
120	1.422592	Apple_d5:84:fc	ITF7-VRRP-VRID_01	0x8000	66	IPv4
121	1.485750	HuaweiTechno_4c:32:1e	Apple_d5:84:fc	0x8000	274	IPv4
122	1.485910	Apple_d5:84:fc	ITF7-VRRP-VRID_01	0x8000	66	IPv4
123	1.490483	Apple_d5:84:fc	ITF7-VRRP-VRID_01	0x8000	651	IPv4
124	1.413788	HuaweiTechno_4c:32:1e	Apple_d5:84:fc	0x8000	814	IPv4
125	1.413947	Apple_d5:84:fc	ITF7-VRRP-VRID_01	0x8000	54	IPv4
126	1.421133	Apple_d5:84:fc	ITF7-VRRP-VRID_01	0x8000	138	IPv4
127	1.423912	HuaweiTechno_4c:32:1e	Apple_d5:84:fc	0x8000	66	IPv4
128	1.476952	HuaweiTechno_4c:32:1e	Apple_d5:84:fc	0x8000	274	IPv4

Packet Details:

- Frame 128: 463 bytes on wire (3784 bits), 463 bytes captured (3784 bits) on interface enb, id 0
 - Section number: 1
 - Interface id: 0 (enb)
 - Encapsulation type: Ethernet (1)
 - Arrival Time: Dec 8, 2023 13:38:09.247699000 HKT
 - UTC Arrival Time: Dec 6, 2023 13:38:09.247699000 UTC
 - Ethernet Arrival Time: 0.000000000 seconds
 - Time shift for this packet: 0.000000000 seconds
 - [Time delta from previous captured frame: 0.000209000 seconds]
 - [Time delta from previous displayed frame: 0.000209000 seconds]
 - [Time since reference or first frame: 1.371252800 seconds]
 - Frame Number: 128
 - Frame Length: 463 bytes (3784 bits)
 - Capture Length: 463 bytes (3784 bits)
 - [Frame is marked: False]
 - [Frame is ignored: False]
 - [Protocol: 0x8000: IEEE 802.3-2005]
- Ethernet II, Src: Apple_d5:84:fc (6c:f7:d5:84:fc), Dst: ITF7-VRRP-VRID_01 (08:00:27:00:00:01)
 - Destination: ITF7-VRRP-VRID_01 (08:00:27:00:00:01)
 - Address: ITF7-VRRP-VRID_01 (00:80:5e:00:01:01)
 -0..... = LG bit: Globally unique address (factory default)
 -0..... = IG bit: Individual address (unicast)
 - Source: Apple_d5:84:fc (6c:f7:d5:84:fc)
 - Address Apple_d5:84:fc (6c:f7:d5:84:fc)
 -0..... = LG bit: Globally unique address (factory default)
 -0..... = IG bit: Individual address (unicast)
- Data (truncated): 450001:100004000040067270a445e48807758c784080585801c642518430f50181000cab0000454207769726573686172662d6c61627327485454820657468657265616c2d6c61622d66696c65332e68746dc2840... (Length: 449)

1. The 48-bit MAC address of the network adaptor on my computer is 6c:7e:67:d5:04:fc
2. The 48-bit destination MAC address in the Ethernet frame is 00:00:5e:00:01:01 This is not the MAC address of gaia.cs.umass.edu but the MAC address of the network adaptor of the router that my computer connected to.
3. The hexadecimal value for the two-byte type field in the Ethernet frame is 0x0800 and the upper layer protocol this corresponds to is IPv4.



5. The source MAC address is 48:d5:39:4c:32:1e and this is the MAC address of the network adaptor of the router that my computer connected to.
6. The destination MAC address is 6c:7e:67:d5:04:fc and this is the MAC address of the network adaptor of my computer.
7. The hexadecimal value for the two-byte type field in the Ethernet frame is 0x0800 and the upper layer protocol this corresponds to is IPv4.

8. There are 54 bytes from the start of the Ethernet frame the ASCII “H” in “HTTP” appear in the Ethernet frame. So, here are $54 + 13 = 67$ bytes from the start of the Ethernet frame the ASCII “O” in “OK” appear in the Ethernet frame.

The image shows a Wireshark packet capture of an HTTP 200 OK response. The packet list on the left shows packet 156 selected, which is a TCP segment from 10.0.0.1 to 10.0.0.2. The packet details pane on the right shows the structure of the TCP segment and the HTTP response. The packet bytes pane on the right shows the raw data in hexadecimal and ASCII. The ASCII pane shows the text "HTTP/1.1 200 OK (text/html)".

Source Port: 80
Destination Port: 63563
[Stream index: 3]
> [Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 1380]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1367621807
[Next Sequence Number: 1381 (relative sequence number)]
Acknowledgment Number: 410 (relative ack number)
Acknowledgment Number (raw): 1476511787
[010] = Header Length: 20 bytes (5)
Flags: 0x010 (ACK)
Window: 237
[Calculated window size: 30336]
[Window size scaling factor: 128]
Checksum: 0xbdc9 (unverified)
[Checksum Status: Unverified]
Urgent Pointer: 0
> [Timestamps]
> [SEQ/ACK analysis]
TCP payload (1380 bytes)
[Reassembled PDU in frame: 160]
TCP segment data (1380 bytes)

0000 00 ed bd 9c 00 00 48 54 54 50 2f 31 2e 31 20 32 HT TP/1.1 2
0001 2c 20 30 3e 20 4d 65 63 20 32 30 32 33 20 31 33 .. 86 Dec 2023 13
0002 3a 33 30 3a 30 39 20 47 4d 54 8d 00 53 65 72 76 138:09 G MT-Serv
0003 68 65 2f 32 2e 34 2e 36 6f 20 69 70 73 20 50 48 .. Apac he/2.4.6
0004 2f 31 2e 30 2e 32 60 2d 60 69 70 73 20 50 48 .. (CentOS) OpenSS
0005 2f 31 2e 30 2e 32 60 2d 60 69 70 73 20 50 48 .. /1.0.2k -fips PH
0006 50 2f 32 2e 34 2e 33 33 20 6d 6f 64 5f 70 65 72 .. P/2.4.33 mod.per
0007 2e 31 30 2e 33 0d 0a 4c 61 73 74 2d 4d 6f 64 69 .. /2.0.11 Perl/VS
0008 66 69 65 64 3a 20 57 65 64 2c 20 30 30 30 32 20 .. 16.3.-L ast-Modi
0009 63 20 32 30 32 31 20 30 36 3a 35 30 30 30 32 20 .. Filed: We d, 06 De
000a 47 4d 54 8d 00 45 54 61 67 3a 20 22 31 31 39 34 .. c 2023 0 6:59:02
000b 2d 36 30 62 64 31 65 32 64 39 30 65 64 31 22 0d .. -68bd1e2 d9ed11"
000c 0a 41 63 63 65 70 74 20 52 61 6e 67 65 73 3a 20 .. [Accept-Range]:
000d 62 79 74 65 73 0d 0a 43 6f 6e 74 65 6e 74 2d 4c .. bytes-C ontent-L
000e 35 30 30 30 30 30 30 30 30 30 30 30 30 30 30 30 .. length: 4 500-Kee
000f 70 2d 41 6c 69 70 65 3a 20 74 69 6d 65 6f 75 74 .. p-Alive: timeout
0010 30 35 2c 20 6d 61 70 30 31 30 30 30 30 43 6f 6e .. -S; max=100-Con
0011 6e 65 63 74 69 61 6e 3a 20 4b 65 65 70 64 6c 6e .. nection: Keep-Al
0012 69 70 65 0d 0a 43 6f 6e 74 65 6e 74 2d 54 73 70 .. ive-Con tent-Typ
0013 68 74 6d 6c 3b 20 68 74 2f 68 74 6d 6c 3b 20 63 .. e: text/ html; ch
0014 61 72 73 65 74 30 30 30 40 20 3d 30 30 30 30 30 .. arset=UTF -8;...
0015 68 74 6d 6c 3b 20 68 74 2f 68 74 6d 6c 3b 20 63 .. html-che ads-cti
0016 74 6c 65 3e 48 69 73 74 6f 72 69 63 61 6c 20 44 .. tleHist orical D
0017 63 2f 31 2e 30 30 30 30 30 30 30 30 30 30 30 30 .. oments :THE BIL
0018 20 47 46 20 52 49 47 48 54 53 3c 2f 74 69 74 74 .. L OF RIG HTS/CI
0019 6c 65 3e 3c 2f 68 65 61 64 3e 0a 0a 3c 62 6f .. le=/hea d-...-bo

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PART II

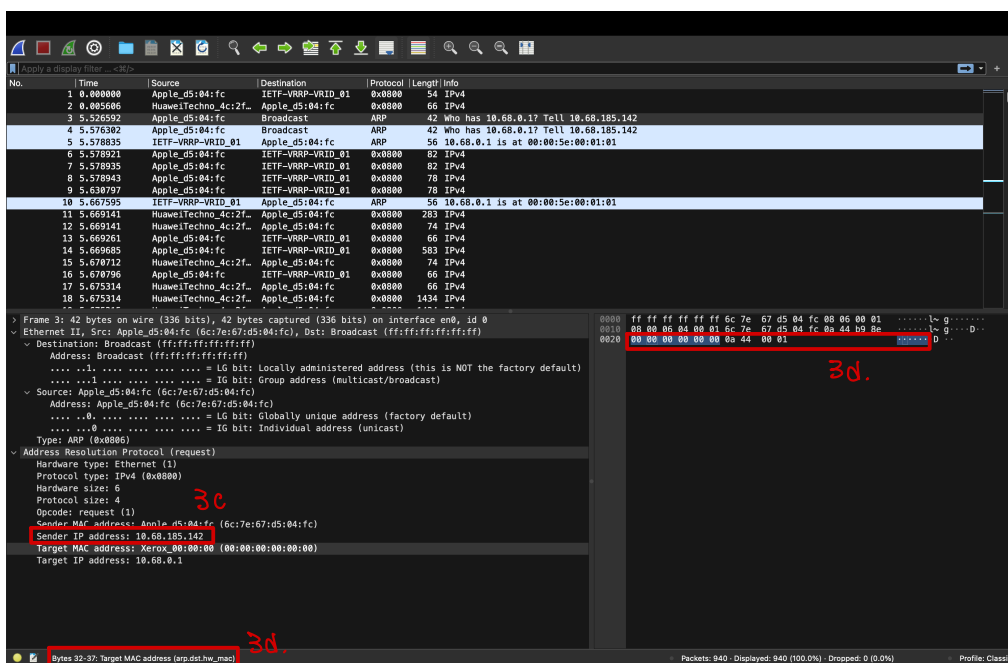
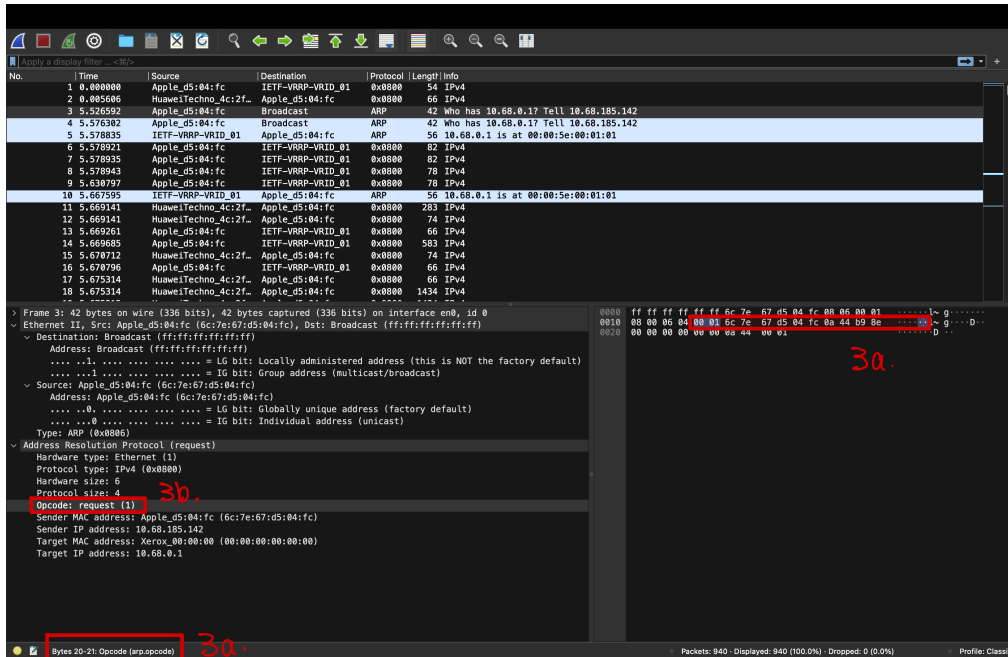
The image shows a Wireshark packet capture of an ARP request. The packet list at the top shows a series of packets, with packet 3 being the ARP request. The packet details pane for packet 3 is expanded, showing the Ethernet II frame and the ARP section. The Ethernet II frame has a destination MAC of ff:ff:ff:ff:ff:ff (Broadcast) and a source MAC of 6c:7e:67:d5:04:fc (Apple_d5:04:fc). The ARP section shows a request for the IP address 10.68.0.1.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Apple_d5:04:fc	IETF-VRRP-VRID_01	0x0000	54	IPv4
2	0.005686	HuaweiTechno_4c:2f...	Apple_d5:04:fc	0x0000	66	IPv4
3	5.576592	Apple_d5:04:fc	Broadcast	ARP	42	Who has 10.68.0.1? Tell 10.68.185.142
4	5.576592	Apple_d5:04:fc	Broadcast	ARP	42	Who has 10.68.0.1? Tell 10.68.185.142
5	5.578835	IETF-VRRP-VRID_01	Apple_d5:04:fc	ARP	56	10.68.0.1 is at 00:00:5e:00:01:01
6	5.578921	Apple_d5:04:fc	IETF-VRRP-VRID_01	0x0000	82	IPv4
7	5.578935	Apple_d5:04:fc	IETF-VRRP-VRID_01	0x0000	82	IPv4
8	5.578943	Apple_d5:04:fc	IETF-VRRP-VRID_01	0x0000	78	IPv4
9	5.630797	Apple_d5:04:fc	IETF-VRRP-VRID_01	0x0000	78	IPv4
10	5.667595	IETF-VRRP-VRID_01	Apple_d5:04:fc	ARP	56	10.68.0.1 is at 00:00:5e:00:01:01
11	5.669141	HuaweiTechno_4c:2f...	Apple_d5:04:fc	0x0000	283	IPv4
12	5.669141	HuaweiTechno_4c:2f...	Apple_d5:04:fc	0x0000	74	IPv4
13	5.669261	Apple_d5:04:fc	IETF-VRRP-VRID_01	0x0000	66	IPv4
14	5.669685	Apple_d5:04:fc	IETF-VRRP-VRID_01	0x0000	583	IPv4
15	5.670712	HuaweiTechno_4c:2f...	Apple_d5:04:fc	0x0000	74	IPv4
16	5.670795	Apple_d5:04:fc	IETF-VRRP-VRID_01	0x0000	66	IPv4
17	5.675314	HuaweiTechno_4c:2f...	Apple_d5:04:fc	0x0000	66	IPv4
18	5.675314	HuaweiTechno_4c:2f...	Apple_d5:04:fc	0x0000	1434	IPv4

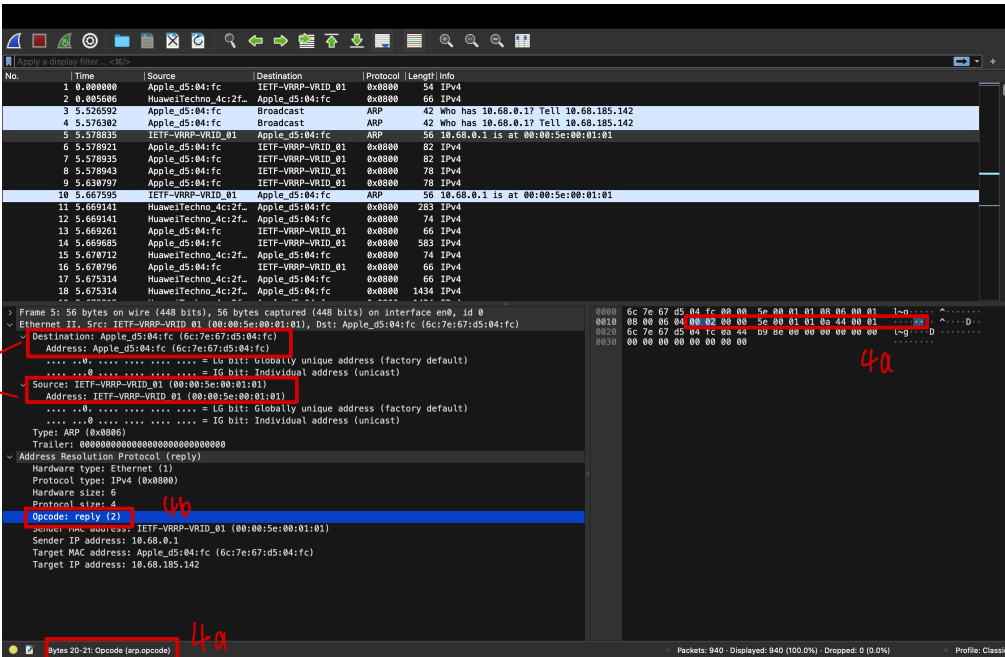
Frame 3: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface en0, id 0
 Ethernet II, Src: Apple_d5:04:fc (6c:7e:67:d5:04:fc), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
 Destination: Broadcast (ff:ff:ff:ff:ff:ff)
 Address: Broadcast (ff:ff:ff:ff:ff:ff)
1. = IG bit: Locally administered address (this is NOT the factory default)
1. = IG bit: Group address (multicast/broadcast)
 Source: Apple_d5:04:fc (6c:7e:67:d5:04:fc)
 Address: Apple_d5:04:fc (6c:7e:67:d5:04:fc)
0. = IG bit: Universally unique address (factory default)
0. = IG bit: Individual address (unicast)
 Type: ARP (0x0806)
 Address resolution protocol (request)
 Hardware type: Ethernet (1)
 Protocol type: IPv4 (0x0800)
 Hardware size: 6
 Protocol size: 4
 Opcode: request (1)
 Sender MAC address: Apple_d5:04:fc (6c:7e:67:d5:04:fc)
 Sender IP address: 10.68.185.142
 Target MAC address: Xerox_00:00:00:00:00:00
 Target IP address: 10.68.0.1

1. The hexadecimal values for the source and destination MAC addresses in the Ethernet frame containing this ARP request message are 6c:7e:67:d5:04:fc and ff:ff:ff:ff:ff:ff respectively.
2. The hexadecimal value for the two-byte type field in the Ethernet frame is 0x0806 and the upper layer protocol this corresponds to is ARP protocol.

3. (a) The ARP opcode field begins from 20 bytes of the beginning of the Ethernet frame.
- (b) The value of the opcode field is 1.
- (c) The ARP message contains the IP address of the sender which is 10.68.185.142
- (d) The “question” in the ARP request message start from 32 bytes of the beginning of the Ethernet frame.

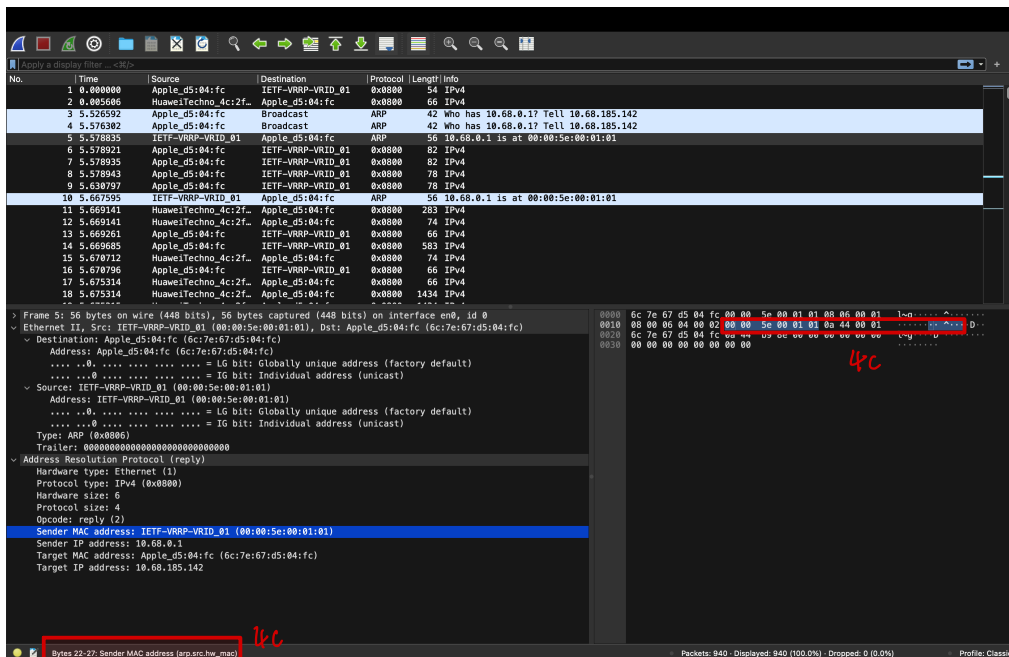


4. (a) The ARP opcode field begins from 20 bytes of the beginning of the Ethernet frame.
- (b) The value of the opcode field is 2.
- (c) The “answer” in the ARP response message start from 22 bytes of the beginning of the Ethernet frame.

5. 

4a

4b



4c

4b

5. The hexadecimal values for the source and destination MAC addresses in the Ethernet frame containing this ARP reply message are 00:00:5e:00:01:01 and 6c:7e:67:d5:04:fc respectively. *(Please refer to the figure in the previous page).*